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MILITARY SPECIFICATION

PUMPING ASSEMBLY, FLAMMABLE LIQUID, BULK TRANSFER, GASOLINE ENGINE DRIVEN, 350 GPM CAPACITY AT 190 FEET TOTAL HEAD, WHEEL MOUNTED

1. SCOPE

1.1 This specification covers a wheel-mounted, gasoline-engine-driven, centrifugal pumping assembly complete with all necessary accessories and connections.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

O-E-915	- Extinguisher, Fire, Dry Chemical (Portable).
QQ-S-781	- Strapping, Steel, Flat and Seals.
MMM-A-187	- Adhesive, Synthetic, Epoxy Resin Base Paste Form, General Purpose.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-P-40	- Packaging and Packing of Hand Tools.

Military

MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.

FSC 4320

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MIL-S-7916	- Sealing Compound, Thread and Gasket, Fuel, Oil and Water Resistant.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-L-21260	- Lubricating Oil, Internal-Combustion Engine, Preservative, and Break-In.
MIL-A-22895	- Adhesive, Metal Identification Plate.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-V-58039	- Valve, Gate, Rising Stem, Double Acting, Aluminum.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U. S. Military Property.
MIL-STD-193	- Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MS35829	- Hinge, Continuous (Piano), .075 Thickness, .187 and .250 Pin Dia.

DRAWINGS

ME

DL13213E9817	- Pumping Assembly, POL, Wheel Mounted, 4-Inch, 350 GPM at 190 Ft. Head.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing, Office, Washington, D. C. 20402.)

HYDRAULIC INSTITUTE

Rating Standards and Test Code Standards.

(Application for copies should be addressed to the Hydraulic Institute, 122 East 42nd Street, New York, N. Y. 10017.)

AMERICAN WELDING SOCIETY

Standard Qualification Procedures.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Welding Qualifications (Section IX, ASME Boiler and Pressure Vessel Code).

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A48	- Gray Iron Castings.
B26-60T	- Aluminum-Base Alloy Sand Castings.
B148-52	- Aluminum Bronze Sand Castings.
A276-60T	- Hot-Rolled and Cold-Finished Corrosion-Resisting Steel Bars.
D2000	- Elastomeric Materials for Automotive Applications.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

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SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, N. Y. 10001.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Tariff Publishing Officer, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street NW, Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Description. The pumping assembly shall be in accordance with Data List 13213E9817 as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish a pumping assembly for examination and testing within the time frame specified to prove prior to starting production that his production methods will produce pumping assemblies that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4).

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3.3 Initial production. When specified, the supplier shall furnish to the Government one or more pumping assemblies for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Material not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4.1 Gaskets. Gaskets shall be fabricated from synthetic rubber conforming to ASTM D2000, type 4BG615A14B14E34F17, except that the discharge check valve gasket shall be fabricated from type 4BG515A14B14E-34F17 synthetic rubber.

3.4.2 Sealing compound. Thread sealing compound for screwed pipe and fittings shall conform to MIL-S-7916.

3.4.3 Threaded fasteners. Threaded fasteners shall be cadmium plated, steel fasteners with threads conforming to Handbook H28.

3.5 Electromagnetic interference. The electromagnetic interference emission and susceptability characteristics of the pumping assembly shall conform to MIL-STD-461, class IIIC.

3.5.1 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

(a) MIL-G-23827 - For instruments, all sealed bearings, and other bearings where operating temperatures do not exceed 250° F.
(b) MIL-G-10924 - For bearings where operating temperatures do not exceed 175° F.

The pumping assembly shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned befor lubrication, because military greases are not always compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.5.2 Oil lubrication. Unless otherwise specified herein, all components shall be filled to the operating level with military oils designated for use in the temperature range as specified herein. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of lubricant used.

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3.6 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose funtions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.7 Engine. One model 4A084 Military Standard 20 horsepower engine will be furnished by the Government for each pumping assembly.

3.8 Pump. The pump shall be of the end suction, top discharge, vertically-split-case centrifugal type and shall be integrally self-priming after initially filling the case. The pump shall prime and deliver full capacity in 1 minute when operated with a static suction lift equivalent to 10 feet of water at standard atmospheric conditions. The pump shall deliver not less than the capacities shown on figure 1. The pump shall require not more than 12.5 feet NPSH at 500 gpm at 170 foot head at 3,600 rpm. The pump shall have minimum efficiencies as follows:

62 percent at 350 gpm at 190 foot head at 3,600 rpm
71 percent at 500 gpm at 170 foot head at 3,600 rpm
72 percent at 500 gpm at 130 foot head at 3,200 rpm

In addition, the pump shall require not more than 23.5 bhp at any operating condition as indicated on figure 1. The suction and discharge ports shall be 4-inch, nominal, flanged. The pump shall withstand an internal hydrostatic pressure of 150 psi without evidence of leakage or permanent deformation.

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3.8.1 Pump case. The pump case shall be aluminum castings conforming to ASTM B26-60T, alloy SG70A-T6. The cutwater shall be a gray iron alloy casting conforming to ASTM A48, class 30 and shall be integrally cast into the pump case. The case shall have mounting feet, provisions for complete drainage, and a 1-1/2 NPT priming port with a 1-1/2 NPT pipe plug. The mounting dimensions of the case shall be such that it is compatible with the intermediate housing and the wheel frame mounting holes.

3.8.2 Impeller. The impeller shall be of the semi-open, nonclogging type, fabricated from aluminum bronze conforming to ASTM B148-52, alloy 9B. The inpeller shall be screwed onto the shaft in a direction opposite to the direction of rotation. The impeller shall be dynamically balanced to a maximum out-of-balance condition of 0.25 inch-ounce at 3,600 rpm.

3.8.3 Shaft sleeve. The pump shaft seal shall be mounted on a shaft sleeve. The shaft sleeve shall be corrosion resistant steel conforming to ASTM A276-60T, type 304.

3.9 Piping and fittings. All piping and fittings on the suction and discharge of the pump shall be flanged and fabricated from aluminum alloy, except as specified herein. The complete piping including valves and couplings shall not overhang the front end or side of the trailer frame by more than 10 inches when installed on the pumping assembly. All piping and fittings shall withstand an internal hydrostatic pressure of 150 psi.

3.9.1 Suction piping. Suction piping shall consist of a suction manifold and a Y-type strainer. The pressure loss through the suction piping, including the gate valves, shall not exceed 11 feet of water at 500 gpm.

3.9.1.1 Suction manifold. The suction manifold shall have one straight run in line with the pump inlet and one branch run parallel to the pump inlet and shall connect the suction valves to the strainer. The manifold shall be such that the suction valves will be inclined 30 degrees downward from the horizontal.

3.9.1.2 Y-type strainer. The strainer shall be a 1/4-inch mesh, reinforced basket strainer fabricated from bronze. The strainer basket shall be removable without interference from or disassembly of other components of the pumping assembly. The strainer inlet and outlet shall be directly in line with the pump inlet.

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3.9.2 Discharge piping. The discharge piping shall consist of a discharge elbow air eliminator, a discharge check valve, and a discharge manifold.

3.9.2.1 Discharge elbow air eliminator. An air eliminator shall be furnished at the highest point in the discharge piping. It shall be an integral part of the pump discharge elbow and shall have either a nonferrous or corrosion-resisting steel mechanism and ball float. The air eliminator shall be capable of eliminating air during the primary cycle of the pump with a totally closed discharge and allowing the pump to prime with a static suction lift equivalent to 10 feet of water in not more than 1 minute. The air eliminator shall have a ball check valve to prevent air from being drawn into the piping through the air eliminator and shall be furnished with a hand operated vent valve and 10 feet of 1/4-inch ID petroleum-resistant rubber drain hose.

3.9.2.2 Discharge check valve. The discharge check valve shall be a lever-operated, spring-loaded, swing check valve located between the air eliminator and the discharge manifold. The check valve shall be mounted in the horizontal position when closed and shall maintain the pump suction vacuum with no loss for not less than 30 minutes.

3.9.2.3 Discharge manifold. The discharge manifold shall consist of a flanged 4-inch, long-turn elbow and a flanged 4-inch Y-pipe. The Y-pipe shall connect the elbow to the two 4-inch discharge valves. The discharge valves shall be located perpendicular to the centerline of the pump and angled 30 degrees downward from the horizontal.

3.9.3 Valves. The piping shall be provided with four aluminum gate valves conforming to MIL-V-58039, size 4-inch, type I. Each valve shall be supported from the wheel frame with not less than one support bracket.

3.10 Grounding assembly. The grounding assembly shall consist of ground rods, ground wires and a ground terminal.

3.10.1 Ground rods. Two ground rods, 48 inches long, each with one end pointed, shall be furnished. The rods shall be fabricated from copper-clad, cold-rolled steel.

3.10.2 Ground wires. Two uninsulated, braided, ground wires, 10 AWG, 6 feet long, shall be furnished. A 50-ampere uninsulated electrical clip shall be soldered to each end of each wire.

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3.10.3 Ground terminal. A ground terminal shall be provided on the wheel frame for attaching the ground wires. The terminal shall be bonded to the frame and located so as not to interfere with the operation of the pumping assembly. The ground terminal shall be clearly identified.

3.11 Toolbox. The pumping assembly shall be provided with a weatherproof toolbox made of 0.075-inch nominal thickness steel. The lid shall be of the overlapping type to prevent entry of rain or dirt and shall be mounted on hinges conforming to MS35829, 0.125-inch diameter brass pin. The box shall have a latch to secure the lid in the closed position and a hasp for locking. The box shall have two drain holes in opposite corners of the bottom. The box shall be mounted to the wheel frame ahead of the battery box in such a position that the lid will open to a minimum of 90 degrees from the horizontal and remain in the open position. The toolbox shall be of sufficient size to hold the battery electrolyte, the pintle hook, and reflectors (where demounted for shipment), the grounding cables and, the operating and maintenance tools specified (see 3.17).

3.12 Identification marking. The pumping assembly shall be identified in accordance with MIL-STD-130. The marking shall be applied to the pumping assembly on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.13 Performance identification plate. A performance identification plate approximately 8-1/2 inches by 11 inches shall be furnished. The chart shall show the complete performance characteristics of the pump when operating at 3,600 rpm and pumping liquid fuel with a specific gravity of 0.75 and when operating at 3,200 rpm and pumping water with a specific gravity of 1.0. All data shall be derived from the actual performance of the preproduction model. The plate shall conform to MIL-P-514, type III, composition C, grade A, class 1. The chart shall be attached to the inside of the toolbox cover with an adhesive conforming to MIL-A-22895, class A, or MMM-A-187.

3.14 Treatment and painting. The portions of the pumping assembly normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, color as specified. Aluminum alloy castings may be cleaned in accordance with MIL-STD-193.

3.15 Government-furnished property. When specified the following property in quantities indicated will be furnished by the Government (see 6.3):

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Item. No.	Description	Identification	Qty. for each pumping assembly
1	Engine, 20 H.P. Mil. Standard.	Model 4A084	1
2	Fire Extinguisher with Vehicle Type Mounting Bracket.	O-E-915 Type III Class 2, Size 2-1/2	1

3.16 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished.

3.17 Items to be furnished with the pumping assembly. No shipment of pumping assemblies shall be made unless repair parts, tools, and accessories are included with the shipment or unless approval for shipment of the pumping assembly without such items has been received from the contracting officer.

3.18 Workmanship.

3.18.1 Castings and forgings. All parts, components, and assemblies of the pumping assembly which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, pits, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

3.18.2 Piping, fittings, hose, cables. All piping, fittings, hose, control cables, and electrical cables shall be supported to prevent damage due to vibration, chafing, or sharp bends.

3.18.3 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.18.4 Welding. The aluminum surfaces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas (argon or helium) shielded-arc method. Welds may be ground, filed, buffed, or chipped but not

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hammered and shall be free from defects such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, porosity, lack of penetration, lack of prescribed fit, slag inclusion, or spatter.

3.18.5 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter other than mill scale that can be removed by chipping and wire brushing. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the pumping assembly to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified, the welding process used in fabrication of the pumping assembly shall be at the option of the supplier.

3.18.6 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society.
Welding Qualifications of the ASME.

Suppliers who only make horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, supplier's standard welder qualification may be substituted in lieu of the above codes provided that the supplier's procedure is equivalent to the above codes. The supplier shall be responsible for determining that automatic welder equipment operators are capable of producing quality welds.

3.18.7 Welding practices.

(a) Preheat of materials being welded, and maximum interpass temperature during welding, shall be in accordance with the supplier's recommendations.

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(b) Tack welds shall be subject to the same quality requirements as final welds.
(c) Work shall be positioned for flat welding whenever practicable.
(d) Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum. When straightening is required, caution shall be exercised to insure that the straightening process does not weaken the part and pumping assembly.

3.18.8 Weld quality.

(a) The supplier shall be responsible for determining the inspections needed to insure that, when the welded parts are assembled together to make the pumping assembly, the pumping assembly shall conform to the inspection requirements specified in section 4.
(b) All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to both the appearance and strength of the weldment.
(c) Undercut in weldments shall be held to a minimum but shall not be more than 1/32-inch deep.

3.18.9 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.18.10 Machine work. Tolerances and gages for metal fits shall conform to the limits specified herein, and on applicable drawings and to the standards of the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Disassembly inspection. Failure of any test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.7 and 6.8.1)

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction pumping assembly shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction pumping assembly shall be tested as specified in 4.6.2.1 through 4.6.2.8. Failure of any test shall be cause for rejection or shall be cause for performing the inspection specified in 4.1.1.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production pumping assemblies will be selected at random by the Government from the pumping assemblies being produced from production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2 through 4.6.2.8 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production pumping assembly shall not exclude the remaining pumping assemblies from the quality conformance inspection and acceptance provisions specified in section 4.

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4.4.1 Inspection failure. Failure of an initial production pumping assembly to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production pumping assemblies and shall be cause for refusal by the Government to continue acceptance of production pumping assemblies until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on pumping assemblies previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all pumping assemblies accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination for minor defects and for tests shall be in accordance with MIL-STD-105, inspection level III.

4.5.2 Examination.

4.5.2.1 Individual. Each pumping assembly shall be examined for the major defects specified in 4.6.1. Presence of one or more major defects shall be cause for rejection.

4.5.2.2 Samples. Samples selected in accordance with 4.5.1 shall be examined for the minor defects specified in 4.6.1. AQL shall be 6.5 percent defective.

4.5.3 Pre-assembly tests.

4.5.3.1 Individual. Each pump and each fuel tank shall be tested as specified in 4.6.2.4.1 and 4.6.2.4.3, respectively. Failure of any test shall be cause for rejection.

4.5.3.2 Samples. Sample impellers and pumps selected in accordance with 4.5.1 shall be tested as specified in 4.6.2.2 and 4.6.2.5. AQL shall be 4.0 percent defective.

4.5.3.3 Shaft and hub. Two specimens from each heat of steel used in the shafts and each heat-treat lot of hubs shall be tested as specified in 4.6.2.3. Failure of the test shall be cause for rejection.

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4.5.4 Assembled unit test. Test each pumping assembly as specified in 4.6.2.9. Failure of the test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. The pumping assembly shall be examined for the following defects:

Major

101. Fabrication not in accordance with drawings.
102. Dimensions not as specified.
103. Assembly of components incorrect or bolted connections not securely tightened.
104. Components missing.
105. Ties, wheels, hubs, axles, and springs not as specified.
106. Fittings, piping, or valves not as specified.
107. Fire extinguisher, bracket, and location not as specified.
108. Toolbox and location not as specified.
109. Performance data plate not as specified.
110. Ground terminal missing, location not as specified, or not bonded to frame.
111. Ground rods and cables not as specified.
112. Piping, fittings, hose, and cable not supported as specified.
113. Welding not as specified.
114. Bearing lubricant not as specified.

Minor

201. Treatment and painting not as specified.
202. Identification marking missing or not legible.
203. Dimensions other than those identified on the SQAP requirements shown on the drawings not as specified.
204. Repair parts and maintenance tools not as specified or missing.

4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed in accordance with the applicable provisions of the test code for centrifugal pumps of the standards of the Hydraulic Institute. Prior to test, the pumping assembly shall be lubricated with greases and oils specified (see 6.6). All performance data shall be corrected to standard sea level conditions of barometric pressure of 29.92 inches of mercury and liquid fuel at 60° F. having a specific gravity of 0.75 and vapor pressure of 4.0 psi. Water at a temperature of between 60° F. to 80° F. may be used as the test fluid.

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4.6.2.2 Balancing. The impeller and the female spline coupling half shall be tested to determine the maximum out-of-balance condition at 3,600 rpm. An out-of-balance condition greater than 0.25 inch-ounce shall constitute failure of this test.

4.6.2.3 Hardness.

4.6.2.3.1 Pump shaft. Test the spline teeth of the shaft for Rockwell C hardness. Readings shall be taken on not less than 4 teeth equally spaced around the spline. Any hardness reading less than Rockwell C 30 or greater than 35 shall constitute failure of this test.

4.6.2.3.2 Spline hub. Test the spline teeth of the hub for Rockwell C hardness. Readings shall be taken on the ends of not less than 4 teeth equally spaced around the spline. Any hardness reading less than Rockwell C 25 or greater than 30 shall constitute failure of this test.

4.6.2.4 Hydrostatic.

4.6.2.4.1 Pump. Apply an internal hydrostatic pressure of 150 psi to the completely assembled pump, with seals in place, for a period of not less than 2 minutes. Any leakage through the pump case or joints shall constitute failure of this test.

4.6.4.2 Pumping assembly. Apply an internal hydrostatic pressure of 150 psi to the completely assembled pump, piping, and fittings for a period of not less than 5 minutes. This test shall be performed with the 4-inch valves completely open and the quick-disconnect fitting covers in place. Any leakage through the pump, fittings, piping, or joints shall constitute failure of this test.

4.6.2.4.3 Fuel tank. Apply an internal air pressure of 5 psig to the fuel tank. All attachments and fuel cap shall be sealed. Immerse in water so that the tank top is not less than 12 inches under the water level. Any evidence of leakage from the tank shall constitute failure of this test.

4.6.2.5 Priming. Fill the pump case with either liquid petroleum fuel or a suitable solvent having a specific gravity of 0.72 to 0.75, and conduct the priming test with a static suction lift equivalent to 10 feet of water at standard atmospheric conditions. This test may be performed lifting either water, gasoline of 0.75 specific gravity, or a solvent of 0.75 specific gravity. Inability of the pump to prime and deliver full capacity in 1 minute shall constitute failure of this test.

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4.6.2.6 Performance. With the pump operating at both 3,200 and 3,600 rpm and a static suction lift equivalent to 10 feet, test the pump in accordance with the Rating Standards and Test Code Standards for centrifugal pumps of the Hydraulic Institute. Data relative to head-capacity shall be taken at a minimum of 6 points and shall include the rating points, wide open discharge, and shutoff. Delivery of less than the capacities shown on figure 1, efficiencies less than those specified in 3.8, an NPSH greater than 12.5 feet at 500 gpm at 3,600 rpm or a horsepower requirement greater than 23.5 bhp. shall constitute failure of this test.

4.6.2.7 Suction piping. With the pumping assembly operating at a minimum suction lift and delivering 500 gpm of liquid, check the pressure loss through the suction piping when using the in-line suction fittings on the manifold. A pressure loss of greater than 11 feet of water shall constitute failure of this test.

4.6.2.8 Operational. Fill the pump case with either petroleum fuel or a suitable solvent having a specific gravity of 0.72 to 0.75 and prime the pumping assembly against a closed discharge with a static suction lift equivalent to 10 feet of water at standard atmospheric conditions. Once the pump has primed, open the discharge and operate the pump at 350 gpm at 190 foot head or 500 gpm at 130 foot head for 100 hours. This test shall be performed by operating for not less than 12 continuous hours daily until the conclusion of the test. During operation, only minor adjustments to the engine and discharge valving may be performed. A maximum shutdown time of 15 minutes is allowable for engine maintenance and examination during each 12-hour period. During operation, determinations shall be made as to whether there is leakage through the pump case, piping, fittings, or at the pump shaft seal and whether there is malfunction of any component. Evidence of leakage, malfunction of any component, inability of the pumping assembly to prime in 1 minute and deliver the capacity specified or inability of the pumping assembly to operate continuously shall constitute failure of this test.

4.6.2.9 Production. Test the pumping assembly as specified in 4.6.2.8 except that the operating time shall be 15 minutes.

4.7 Inspection comparison. When initial production inspection is specified, the Government may select pumping assemblies at any time during the contract production period and subject these pumping assemblies to the examination specified in 4.6.1 and the tests specified in 4.6.2 to determine that the quality of the selected pumping assemblies is equal to the quality standards established during initial production inspection. The inspection will be performed by the Government at a site selected by the Government. Pumping assemblies will

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be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model(s) and initial production pumping assemblies.

4.7.1 Inspection failure. Failure of any pumping assemblies when inspected as specified in 4.7 may be considered cause for refusal by the Government to continue acceptance of pumping assemblies until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.8 Inspection of preparation for delivery.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. Examine the preproduction pack for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completely processed pumping assembly prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

115.	Materials, methods, and containers not as specified for level A. Each incorrect material, method, or container shall be considered one defect.
116.	Disassembly not as specified for level A.
117.	Surfaces requiring a contact preservative in accordance with the referenced document not coated with preservative for level A.
118.	Pintle hook and reflectors not placed in the toolbox for level A.
119.	Interior surfaces of the pump and connecting piping not coated with oil as specified for level A.
120.	Excess oil not drained for level A.
121.	Valves not closed for level A.

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122. Caps and plugs not installed on fittings for level A.
123. Engine components not preserved as specified for level A or C.
124. Tires not inflated to 10 pounds above pressure value for level A.
125. Fire extinguisher not secured in mounting bracket for level A.
126. Components not placed in toolbox as specified for level A.
127. Items not cushioned in toolbox as required for level A.
128. Toolbox lid not closed and secured for level A or C.
129. Packing for level A not in accordance with figure 2.
130. Marking illegible, incomplete or incorrect for level A or C.
131. Special marking not applied as specified for level A or C.

5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination within the time frame specified to prove prior to starting production packaging that the applied preservation, packaging, packing, and marking comply with the preparation for delivery requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.5). The preproduction pack may be accomplished utilizing either the preproduction pumping assembly or a production pumping assembly. If the preproduction model pumping assembly is utilized, any preservation, packaging, and packing shall be removed by the supplier at no expense to the Government when requested by the Government to facilitate comparison between the preproduction model pumping assembly and the production pumping assembly.

5.2 Preservation and packaging. Preservation and packaging shall be level A or C, as specified.

5.2.1 Level A.

5.2.1.1 Disassembly. The pintle hook and reflectors shall be removed from the frame and shall be placed in the toolbox. The screws and lockwashers shall be reinstalled on the pintle hook and reflectors to prevent loss.

5.2.1.2 Unprotected surfaces. Unpainted exterior metal surfaces of components requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

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5.2.1.3 Pump. Interior surfaces of the pump and connecting piping shall be coated with lubricating oil conforming to MIL-L-21260, type I, grade 30. The pump shall be actuated during application of the oil to insure coating of all interior surfaces. After interior preservation is completed, the excess oil shall be drained. The valves shall be closed. Openings into the crossover and Y-type fittings shall be closed with the caps and plugs provided. Exterior surfaces of the pump requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.2.1.4 Engine. The engine and engine accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, as specified for type II, method I.

5.2.1.5 Tires. Tires shall be inflated to 10 pounds above pressure value recommended for maximum load.

5.2.1.6 Fire extinguisher. Unpainted surfaces of the fire extinguisher requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. The fire extinguisher shall be placed in the mounting bracket and secured.

5.2.1.7 Ground rods and cable. The ground rods shall be secured in place on the frame. The ground cable shall be coiled and placed in the toolbox.

5.2.1.8 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts, and the repair parts placed in the toolbox.

5.2.1.9 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with the level A requirements of PPP-P-40, and placed in the toolbox.

5.2.1.10 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3, and placed in the toolbox.

5.2.1.11 Toolbox. The items contained in the toolbox shall be cushioned in a manner to prevent damage to the items. The toolbox lid shall be closed and secured.

5.2.2 Level C.

5.2.2.1 Engine. The engine shall be preserved in accordance with the level C alternate procedure requirements of MIL-E-10062.

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5.2.2.2 Repair parts, maintenance tools, and technical publications. Repair parts and technical publications shall be preserved and packaged as specified for level A. Maintenance tools shall not require preservation and packaging but shall be placed in the toolbox.

5.2.2.3 Toolbox. The toolbox lid shall be closed and secured.

5.2.2.4 Other components. Other components of the pumping assembly not specifically mentioned in 5.2.2.1, 5.2.2.2, or 5.2.2.3, shall be preserved and packaged to afford protection against deterioration and damage from the supplier to the initial destination. The supplier may use his standard commercial practice when it fulfills these requirements.

5.3 Packing. Packing shall be level A or C as specified.

5.3.1 Level A. The wheel-mounted pumping assembly shall be prepared for mobile shipment. A nominal 4-inch by 4-inch wood member shall be secured to the rear frame member, shall extend the full width of the frame, and shall be flush with the top of the frame. The wood member shall be secured with four 1/4-inch by 4-1/2-inch carriage bolts placed through the existing reflector mounting holes. The assembly shall be protected with a cap as shown on figure 2. Flat steel strapping shall conform to QQ-S-781. The two center straps shall completely encircle the cap and shall be applied beneath the drawbar tube under the frame. The front strap shall completely encircle the cap and shall be applied under the lunette. The rear strap shall completely encircle the cap and shall be applied around and under the wood member previously attached to the frame. The valve boxes shall be of such length to be completely free from the tires and the lunette and shall be secured to the cap by nailing.

5.3.2 Level C. The wheel-mounted pumping assembly shall be prepared for mobile shipment in a manner to assure carrier acceptance and safe delivery at lowest rating in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking.

5.4.1 For shipment. Packages and capped and uncapped mobile units shall be marked in accordance with MIL-STD-129.

5.4.2 Special marking. Each pumping assembly shall be marked with 1/2-inch stenciled letters as follows:

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CARLOADING INSTRUCTIONS

Keep frame level.
Block under rear.
Rest front on support leg.
Tie down front and rear.
Block and tie wheels.

For level A the location shall be as indicated on figure 2. For level C the marking shall be on 1/4-inch plywood panels attached to both front and rear of the pumping assembly. For level A the location shall be as indicated on figure 2. For level C the marking shall be on 1/4-inch plywood panels attached to both front and rear of the pumping assembly.

6. NOTES

6.1 Intended use. The pumping assembly covered by this specification is a component of the Fuel System, Supply Point: Portable, 60,000 Gallon Capacity.

6.2 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2).
(c) When initial production inspection is required (see 4.3).
(d) Applicable list of repair parts and maintenance tools (see 3.16).
(e) Time frame required for submission of the preproduction pack (see 5.1).
(f) Level of preservation and packaging and level of packing required (see 5.2 and 5.3).

6.3 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.15.

6.4 Preproduction model. Any changes or deviations of production pumping assemblies from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish pumping assemblies conforming to this specification.

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6.5 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack, and mark the pumping assembly in accordance with this specification.

6.6 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the pumping assembly covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.7 Supplementary Quality Assurance Provisions (SQAP). The contracting officer should require the supplier to maintain records of all SQAP inspections. A suggested paragraph is as follows:

> "The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.8 Definitions.

6.8.1 Supplementary Quality Assurance Provisions (SQAP). A SQAP is a contractual requirement that supplements section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

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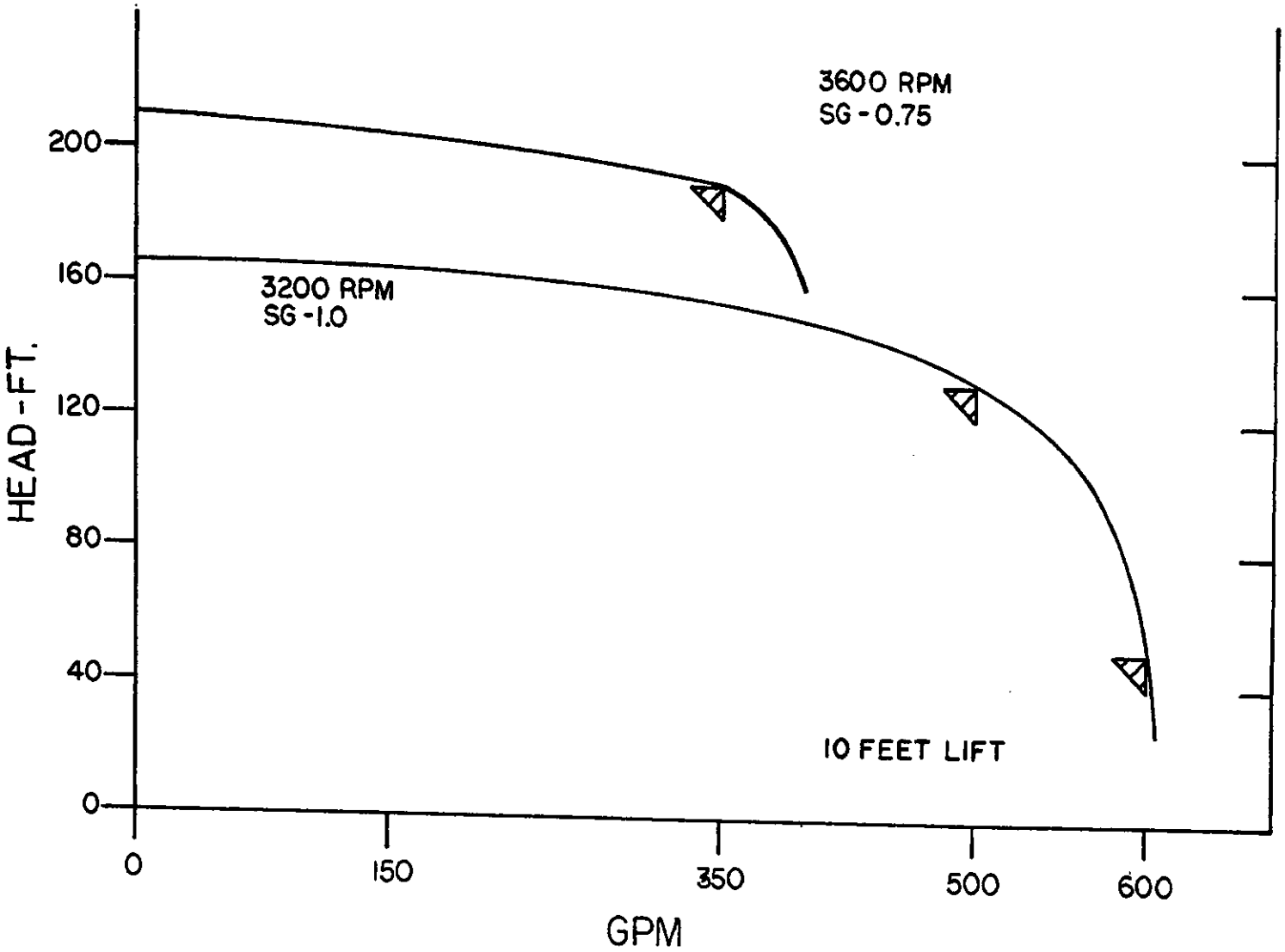
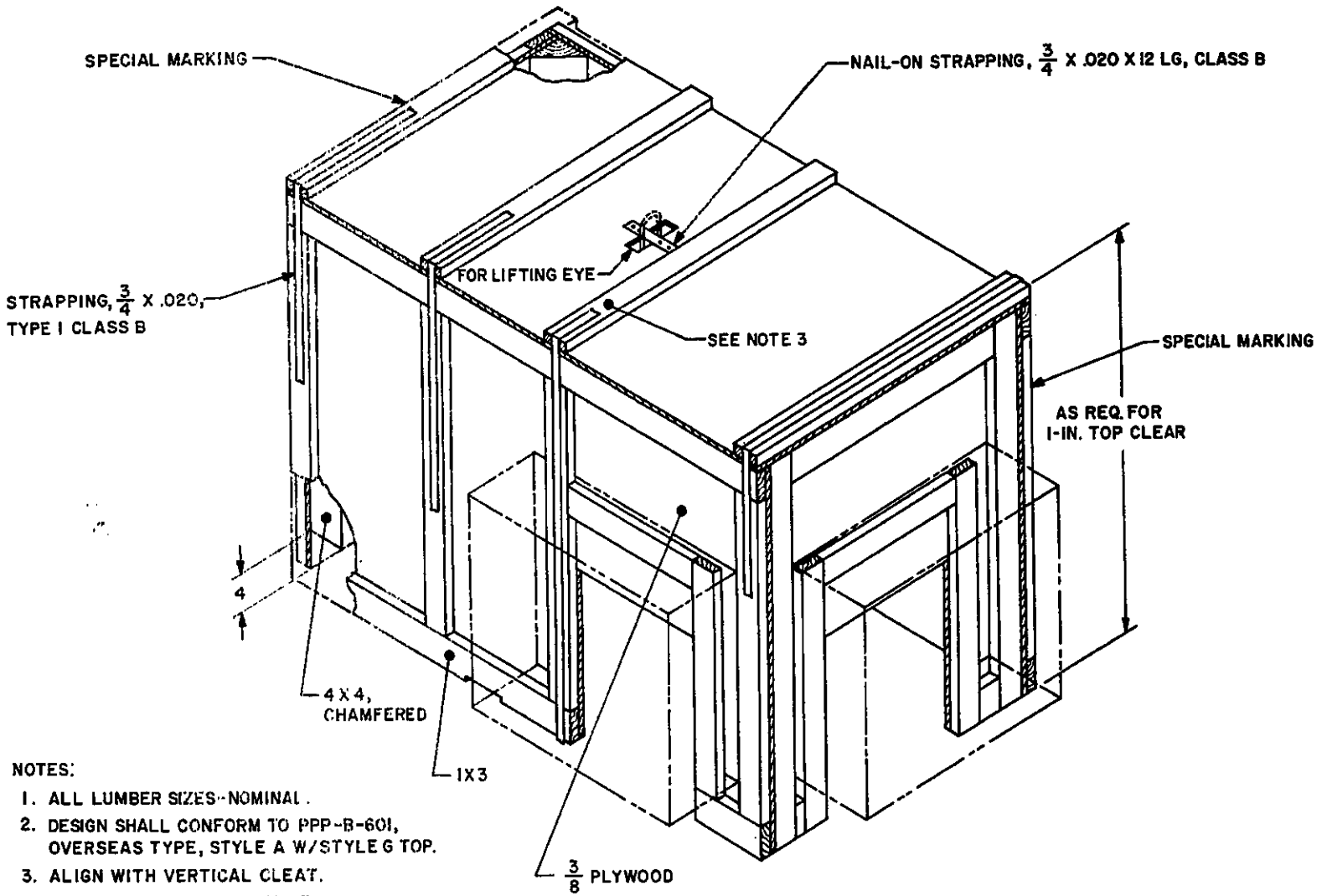


FIGURE 1 PUMP PERFORMANCE CURVES CX299 A

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NOTES:

1. ALL LUMBER SIZES--NOMINAL.
2. DESIGN SHALL CONFORM TO PPP-B-601, OVERSEAS TYPE, STYLE A W/STYLE G TOP.
3. ALIGN WITH VERTICAL CLEAT.
4. NOTCH SIDES TO CLEAR AXLE.
5. ADD 4 SIDES, STYLE A BOXES TO COVER VALVES. NO TOP CLEATS. BOXES TO CLEAR LUNETTE AND NOT EXCEED TIRE WIDTH.

FIGURE 2. PACKING DETAIL

X-2132

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